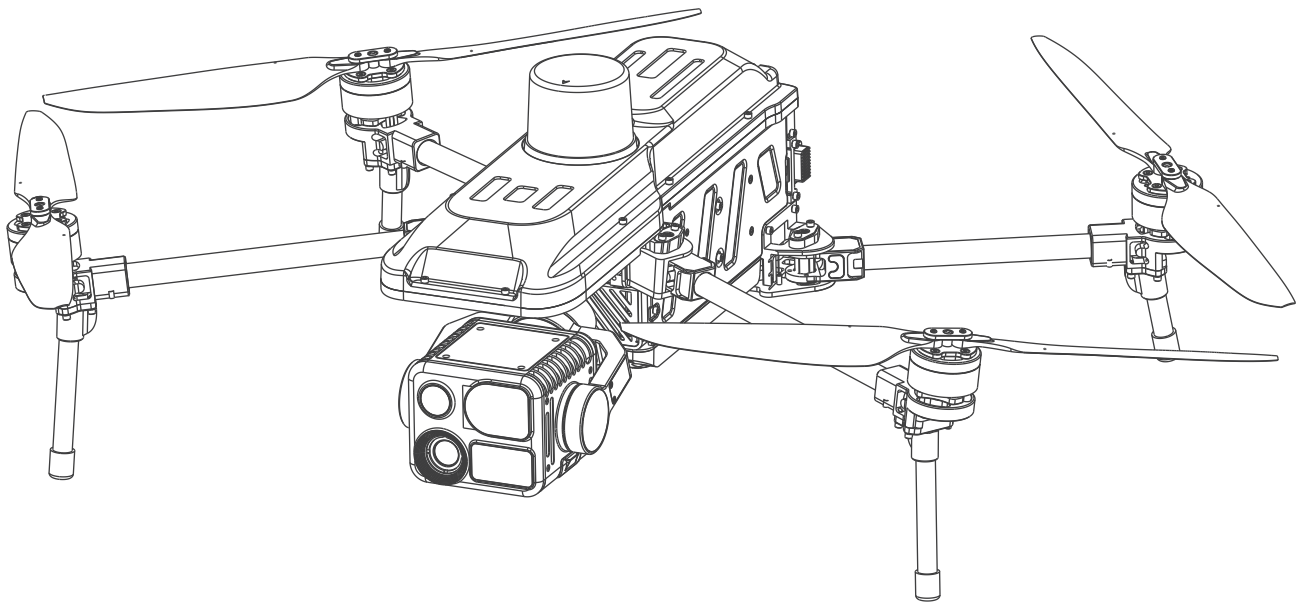


NOVA-4T

Quick Start Guide and Disclaimer

V1.0 202508



Safety Operation Guidelines

Please read the user manual carefully for the correct use of NOVA-4T. Before use, please thoroughly understand and comply with local laws and regulations. Once you use this product, it is deemed that you have carefully read the disclaimers and warnings, and understood, acknowledged, and accepted all terms and content of this statement. You agree to bear full responsibility for the use of this product and any potential consequences that may arise.

1. Never approach rotating propellers and motors.
2. If the aircraft is involved in an accident (such as a collision, overturn, etc.) or exhibits abnormal flight status, the user must immediately stop using it.
3. During flight, if the GCS Fly APP displays a low battery warning, follow the guidance provided, return and land promptly. Do not continue flying forcibly.
4. After landing, ensure the motors have completely stopped before turning off the aircraft and remote controller power. Avoid touching the motor surfaces to prevent burns.
5. Do not modify the aircraft or its related components yourself, as this may affect performance and even cause accidents.
6. Never power on the aircraft if it has gotten wet, as this will cause permanent damage.
7. When upgrading firmware, calibrating the system, or setting parameters, always keep the aircraft away from crowds or animals.

Drone Flight Safety Restrictions

To ensure public safety and maintain order in the airspace, please adhere strictly to the following flight restrictions:

1. Prohibited operation while impaired

Operating drones is strictly prohibited while under the influence of alcohol or drugs, or when experiencing dizziness, fatigue, or other conditions that may impair judgment or control.

2. Prohibited in unauthorized no-fly zones

Flying drones is strictly prohibited in unauthorized airspace over or near sensitive areas, such as airports, train stations, power plants, government buildings, and military facilities.

3. Prohibited from Carrying Hazardous or Prohibited Items

It is strictly forbidden to use drones to transport or drop any flammable, explosive, toxic, or other dangerous items that violate laws and regulations.

4. Prohibition on Exceeding Altitude Limits

Flight altitudes must adhere strictly to local laws, regulations, and restrictions imposed by the aviation authority. Exceeding permitted altitudes is strictly prohibited.

5. Prohibition on Violating Privacy

The illegal tracking, surveillance, or filming of private residences, vehicles, or activities is strictly prohibited. Respect and protect personal privacy rights at all times.

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Product Overview

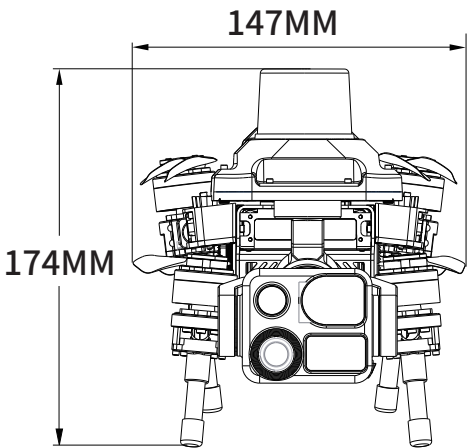
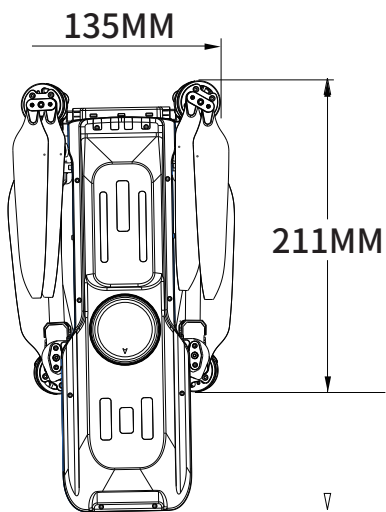
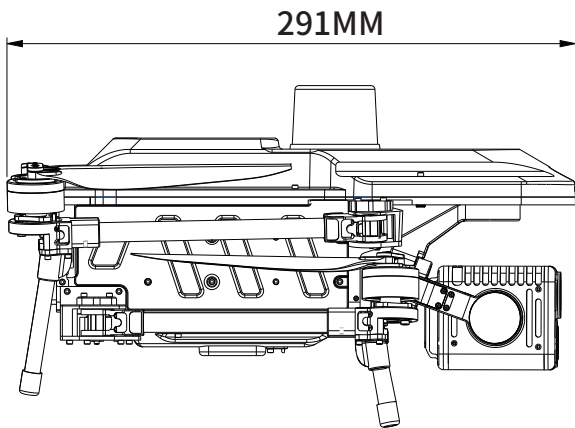
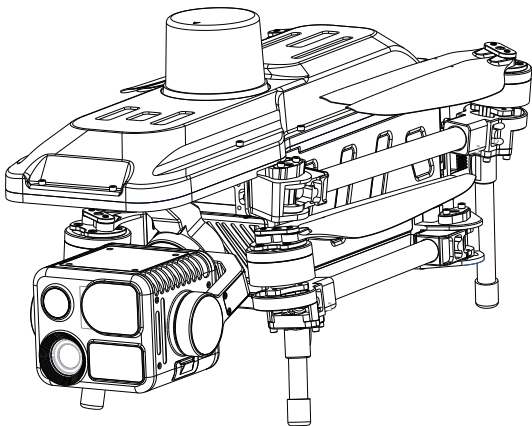
The NOVA-4T is a professional-grade quadcopter drone with a carbon fiber and composite plastic structure that provides high rigidity and a lightweight design. Its foldable design makes it portable and easy to deploy, ideal for tasks such as aerial photography and surveying. With robust power performance, the drone can withstand winds up to Force 6 and deliver extended endurance for prolonged operations. It is equipped with a high-precision multi-mode positioning system that maintains stable flight with a hovering accuracy of ± 0.1 meters. It features a 48-megapixel wide-angle lens, a 48-megapixel telephoto lens, and an infrared thermal imager complemented by a laser rangefinder that measures distances from 5 to 1,200 meters, enabling nighttime reconnaissance and long-range monitoring. The standard screen-based remote controller offers a control range of 10–12.5 kilometers. This industry-grade drone is widely applicable in professional scenarios such as agriculture, security, and emergency rescue due to its outstanding performance and comprehensive functionality.

Parameter List

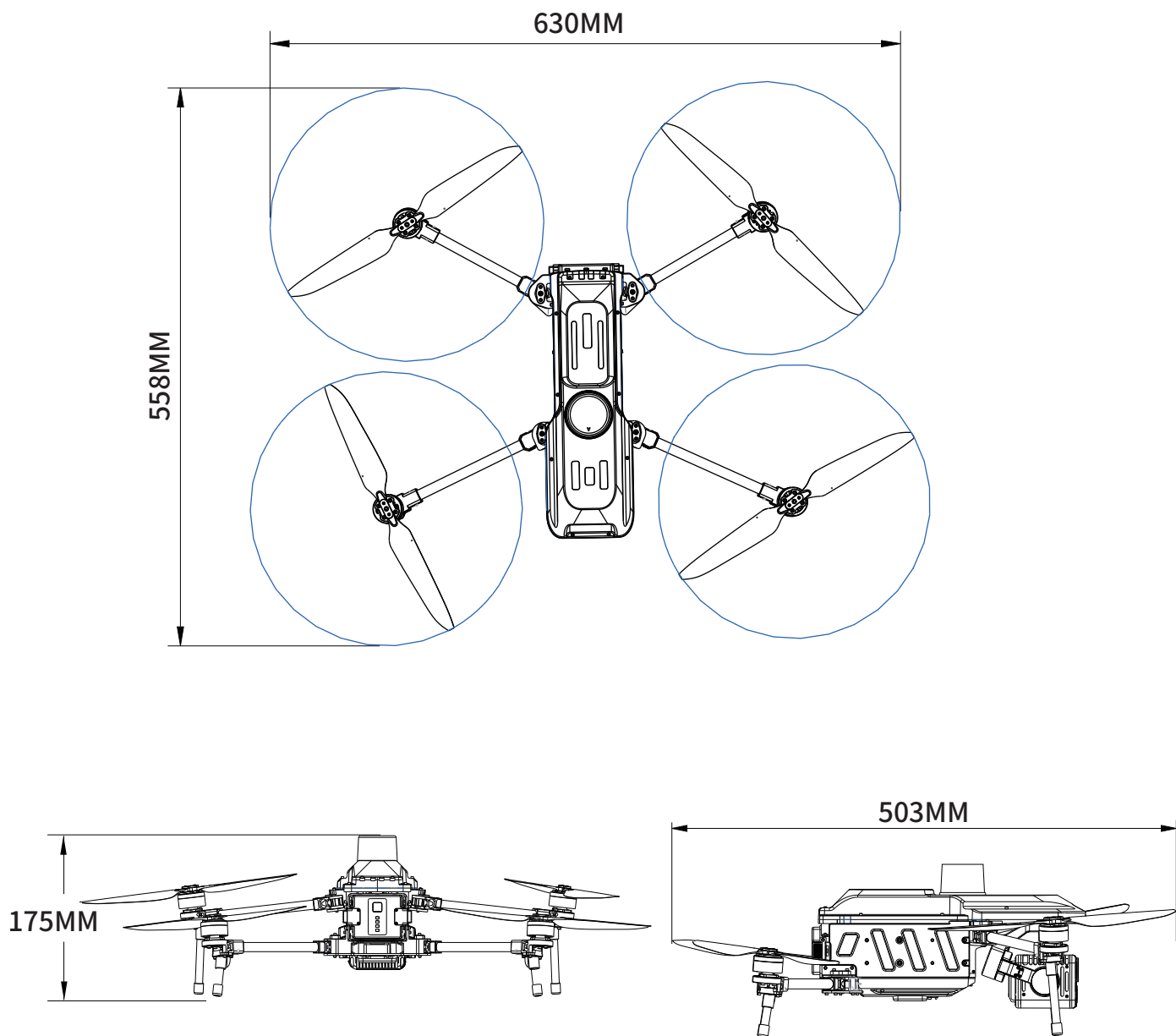
Drone Parameters			
Type	Quadcopter	Maximum Flight Time	45min-52min
Wheelbase	480mm	Recommended Cruising Speed	8-15m/s
Takeoff Weight	1730g	Maximum Horizontal Flight Speed	20m/s
Maximum Load	300g	Maximum Cruising Range	25km （9m/s）
Recommended Load	300g	GNSS Positioning Type	GPS+Galileo+BeiDou+GLONASS
Main Material	Carbon fiber + plastic	Maximum Take-Off Altitude	4000m
Control Radius	10-12.5km (without interference and obstruction)	Maximum Ascent Speed	6m/s
Mission Flight Altitude	Recommended less than 1000m	Maximum Descent Speed	4m/s
Wind Resistance	level 6	Unfolded Size	661X583X175mm
Hovering Time	45min-47min	Folding Method (with blades)	Collapsible
Positioning Method	GNSS+Optical Flow+TOF	Target Solution	Supported
Hovering Accuracy	Vertical: +/-0.1m (visual positioning), +/-1m (GNSS positioning) Horizontal: +/-0.3m (visual positioning), +/-1m (GNSS positioning)		
Communication Frequency Band	Standard 2.4G/5.8G (other frequency bands can be customized)		
Standard Configuration	1*Main frame, 1*K40T four-sensor AI gimbal camera, 1*remote control with screen, propeller*2 sets, 1*portable packaging box		
Battery capacity	4S 12500mAh		

Product Dimensions (unit: mm)

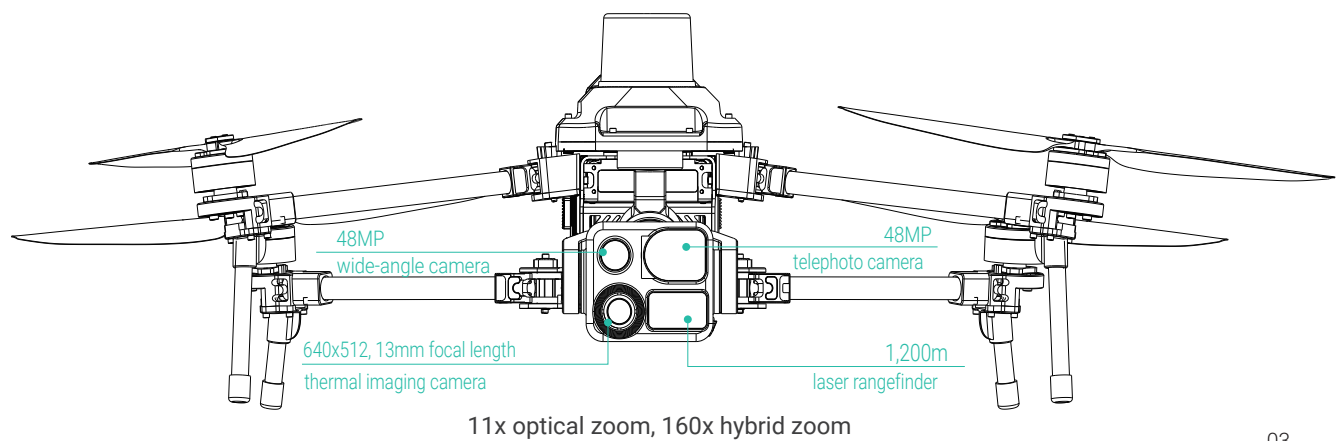
Folded Dimensions



UNFOLDED DIMENSIONS



K40T GIMBAL



PARAMETER LIST

Gimbal Parameters	
Axes	Mechanical three-axis
Control Angle Range	Pitch: -135° to 45° Roll: -45° to 45° Yaw: -135° to 135°
Maximum Control Speed	180°/S
Angle Jitter	±0.005°
Dimensions	114x84x95mm
Interface	Ethernet port/serial port/sbus

Infrared Camera Parameters	
Detector Type	Vanadium Oxide Uncooled Infrared Focal Plane Detector
Resolution	640x512 13mm focal length
Pixel Pitch	12μm
Frame Rate	50Hz / 25Hz
Response Band	8~14μm

Wide Angle Camera	
Sensor size	1/2 inch
Effective Pixels	48 million
FOV	ME FOV:85.4° DFOV:84°±2°
Wide-angle Focal Length	4.49mm
Wide-angle Aperture	F2.8

Zoom Camera	
Sensor size	1/2 inch
Effective Pixels	48 million
Variable aperture	F3.7 ~F4.6
Effective Focal Length	15.2~50mm

Laser Rangefinder	
Measurement range	5-1200 m
Measurement accuracy	±1 m

REMOTE CONTROL PARAMETERS

Image Transmission		Battery	
Antenna	Dual Antenna	Battery Capacity	5000mAh
Maximum Effective Signal Range	15km	Charging Method	PD 30W Fast Charging
Operating Frequency	7+7(2.4G+5.8G)	Charging Time	1.5 hours (powered off)
Equivalent Isotropically Radiated Power (EIRP)	15dBm~29dBm	Operating Time	4.5 hours (power adaptive mode, room temperature 25°C)
Bluetooth		WIFI	
Bluetooth Protocol	5.0	WiFi Protocol	IEEE 802.11a/b/g/n/ac
Operating Frequency	2.4G	Wi-Fi EIRP	17dBm (max)
Bluetooth EIRP	8dBm (max)	Operating Frequency	2.4G/5.8G
Display			
Resolution	1920x1080	Frame Rate	60FPS
Size	5.5 inches	Max Brightness	1000nit
Touch	5-point multi-touch		
General Parameters			
System	Android 11		
HDMI	1080p		
Protection level	IP53		
Weight	700g		
Hardware Interfaces	Charging: Type-C, External Storage: TF card slot		
Operating Temperature	-10℃~45℃		
Memory	4GB RAM, 32GB internal storage (expandable via Micro SD card)		
Dimensions	Length 175mm*Width 137mm*Thickness 47.4mm (without joystick installed) Length 175mm*Width 137mm*Thickness 64mm (with joystick installed)		

RECEIVER PARAMETERS

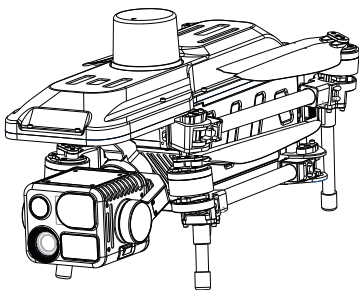


Hardware Interface	SBus:3-Pin	Data Transmission (connect flight control):Uart4-Pin	
	Network Interface:4-Pin	USB Interface:4-Pin	
Remote Control Signal	16 channels SBus	Weight (without antenna)	50 g
Antenna Gain	3 dBi	Working Voltage	12V-27V
Working Environment Temperature	-10℃ ~ 45℃	Size (without antenna)	57 x 47 x 18.1(mm)

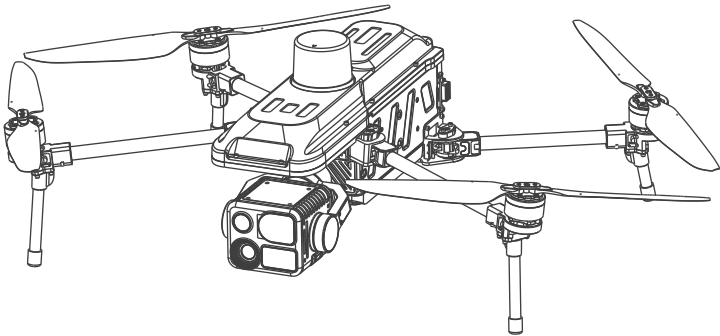
GETTING STARTED

The NOVA-4T folds by default. Follow these steps to prepare the drone and remote controller.

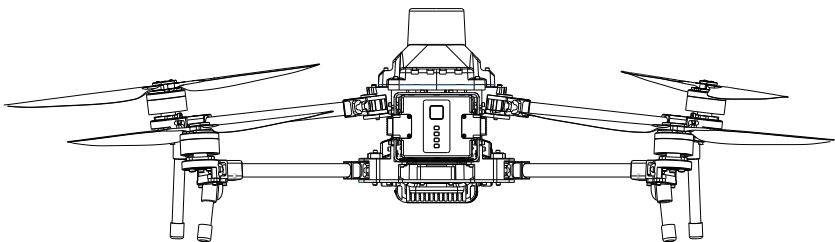
1. Unfold the arms



2. Unfold the propellers



3. Install the smart battery. Ensure that it is properly seated, then briefly press the battery power button once. Next, hold it down to power on the drone.



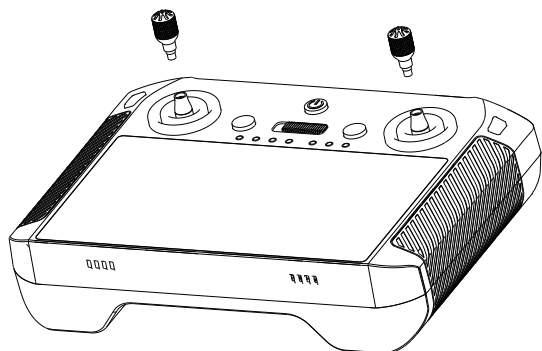
 **Warning**

- 1. Always deploy the front arms before the rear arms.
- 2. Ensure that all arms and propellers are fully extended before powering on the drone, in order to avoid interfering with the gimbal self-check.
- 3. Verify that the smart battery is securely installed before powering on.

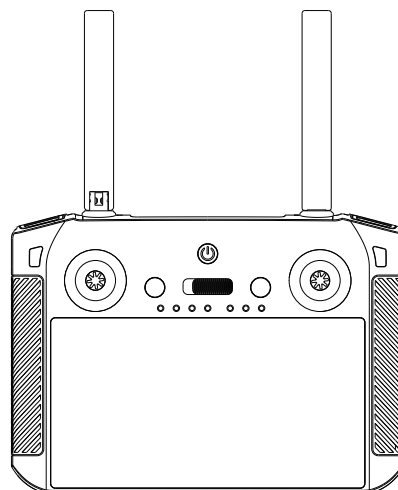
Prepare the Remote Control

Follow these steps to prepare the SCEPTER15 remote control.

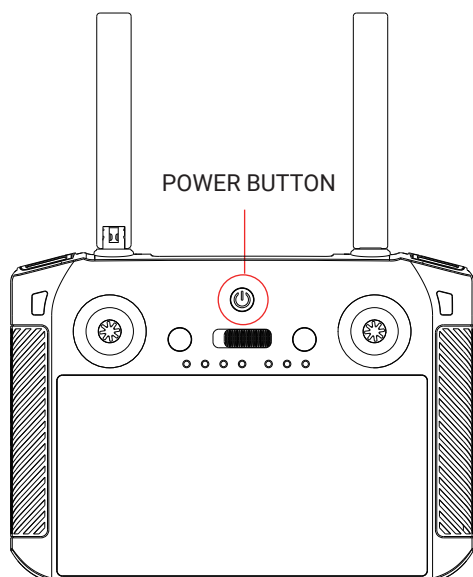
1. Remove the joystick from its slot and attach it to the remote control.



2. Extend the antenna.



3. Briefly press the power button, then hold it down to turn on the remote control.

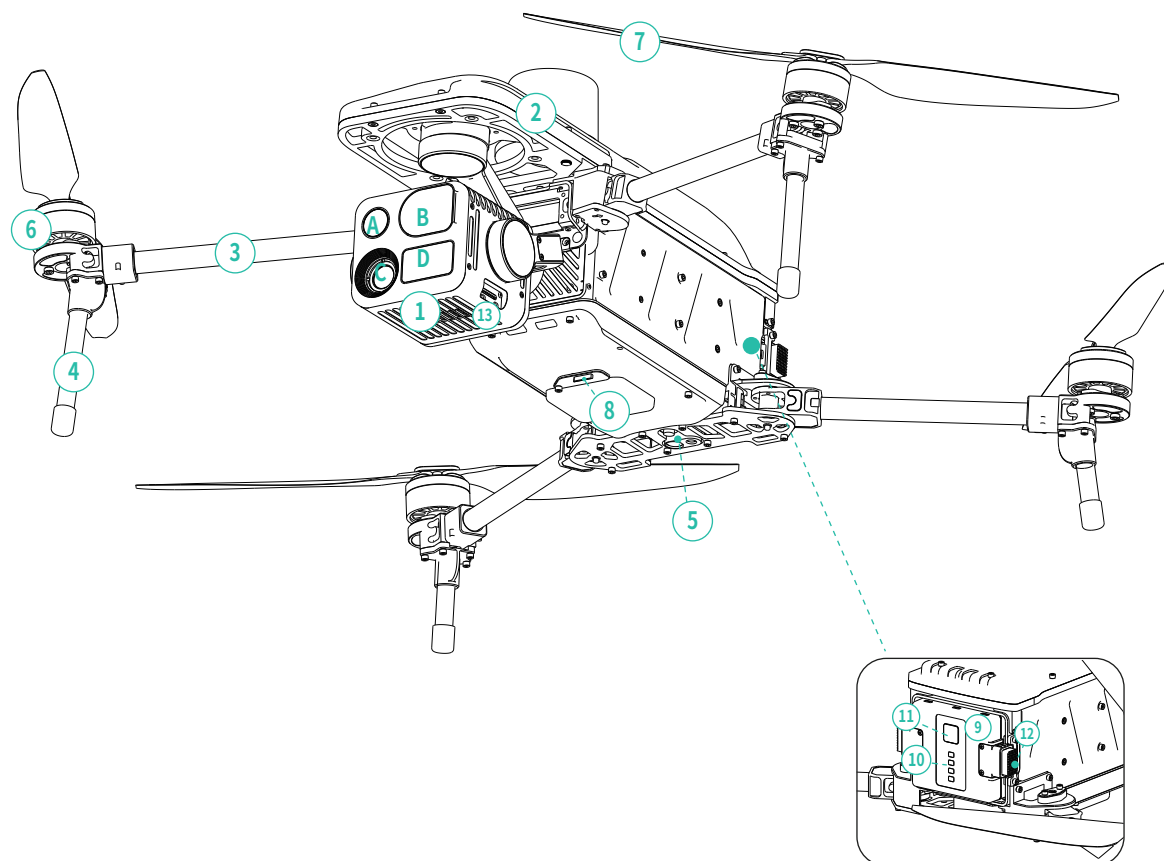


Warning

Make sure the joystick is securely fastened before takeoff.

Component Names

NOVA 4T



1.K40T Gimbal

A. Wide-Angle Camera

B. Telephoto Camera

C. Thermal Imaging Camera

D. Laser Rangefinder

2.GPS

3. Arm

4. Tripod with integrated antenna

5. Optical Flow Module

6. Motor

7. Propeller

8. Flight controller Type-C port

9. Intelligent Battery

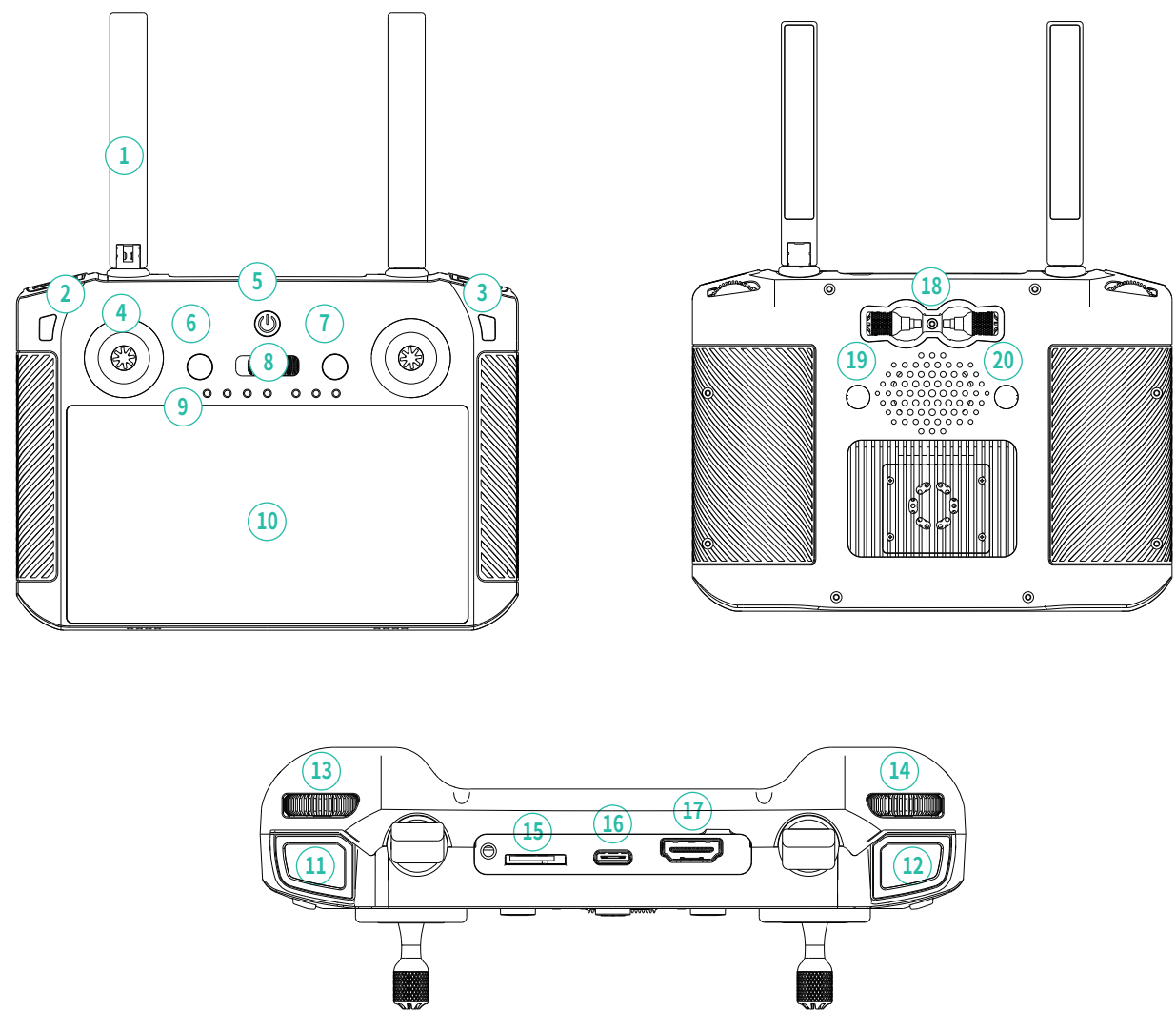
10. Battery indicator light

11. Battery Switch

12. Battery Latch

13. Gimbal SD card slot

SCEPTER15 Remote Controller

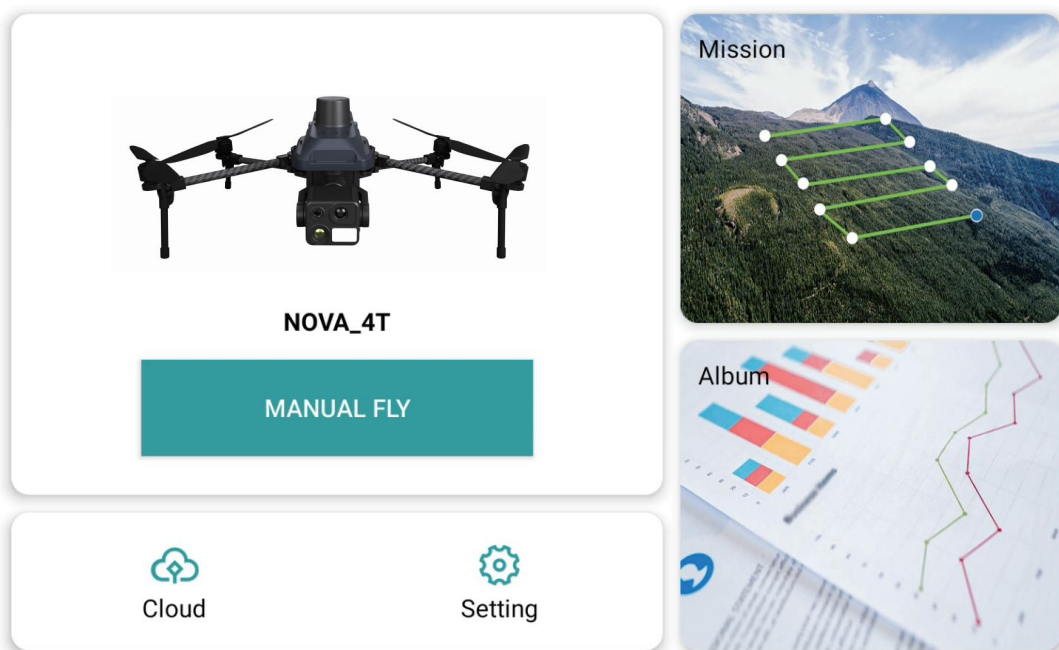


- | | | |
|------------------------------------|--|---------------------------|
| 1. Antenna | 8. Flight Mode Switch Button | 15. SD card slot |
| 2. Streaming button | 9. Battery Indicator | 16. Type-C Charging Port |
| 3. Gimbal Center Return | 10. Touchscreen display | 17. HDMI port |
| 4. Joystick | 11. Photo Button | 18. Joystick Storage Slot |
| 5. Power button | 12. Video Recording Button
(Start/Stop Recording) | 19. R3 Custom Button |
| 6. C1 Custom Button | 13. Gimbal Pitch Wheel | 20. L3 Custom Button |
| 7. One-Touch Return-to-Home Button | 14. Gimbal Yaw Wheel | |

GCS FLY APP

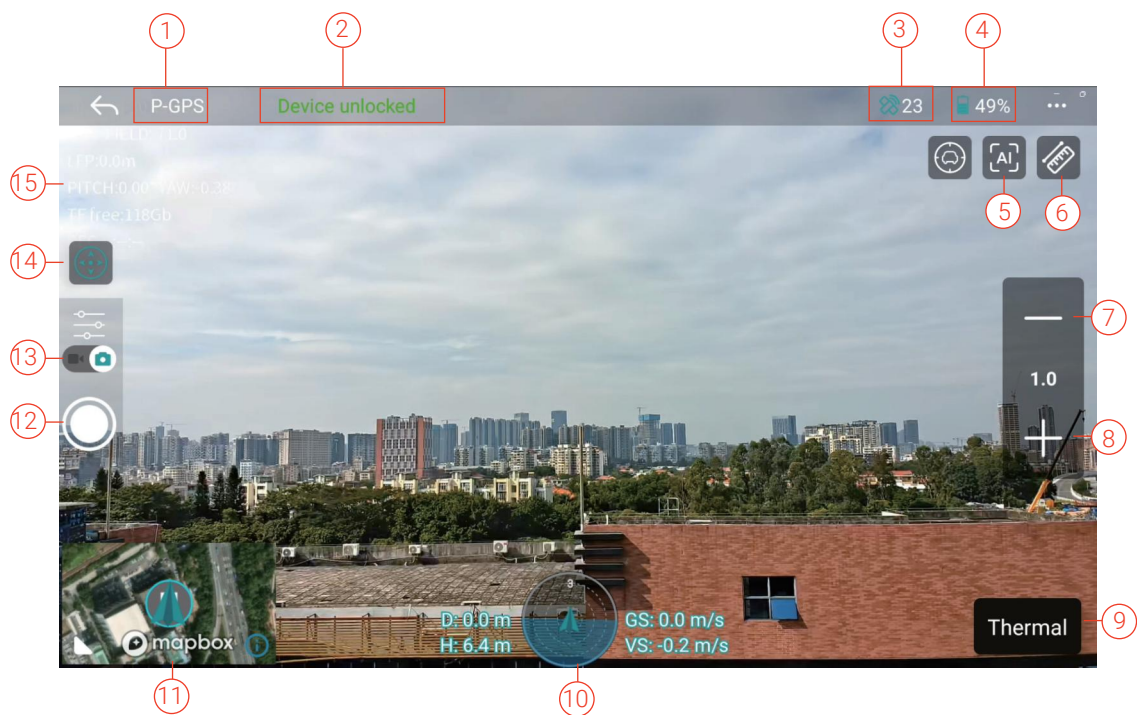
Home

Launch the GCS Fly app to access the home screen.

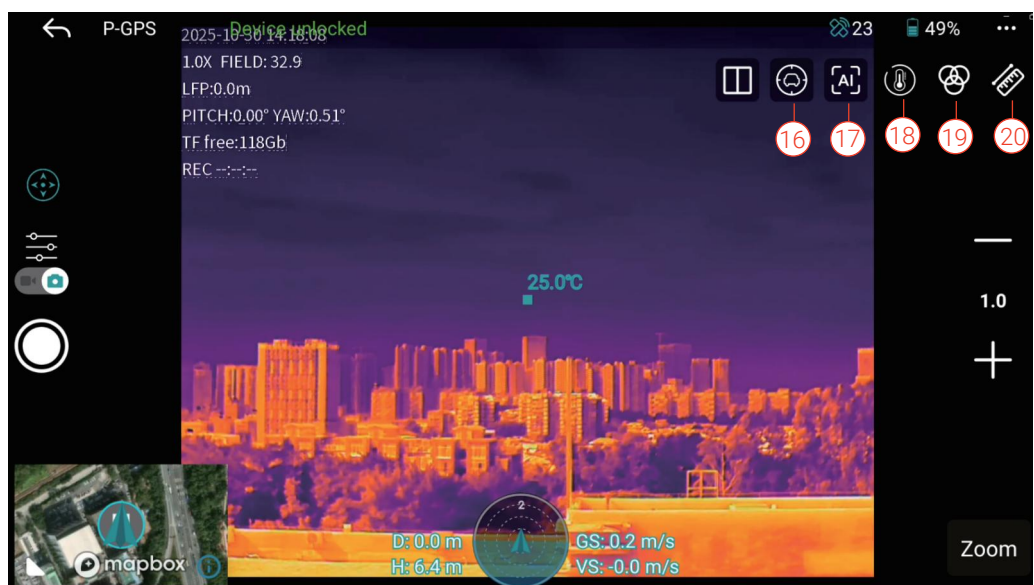


- Settings
Map switching, language settings, and connection settings.
Flight Tasks
- Waypoint editing and uploading, executing flight tasks.
Gallery
- Access the K40T gallery.

Camera Interface



- | | | |
|---------------------------|----------------------------|--------------------------------------|
| 1. Aircraft Gear Position | 6. Laser Rangefinder | 11. Map |
| 2. Aircraft status bar | 7. Zoom Out | 12. Photo/video capture |
| 3. GPS satellite count | 8. Zoom in | 13. Switch to Photo/Video Capture |
| 4. Aircraft battery level | 9. IR Switch | 14. Gimbal One-Touch Center/Down Key |
| 5. Gimbal tracking | 10. Aircraft Attitude Ball | 15. Gimbal OSD target resolution |



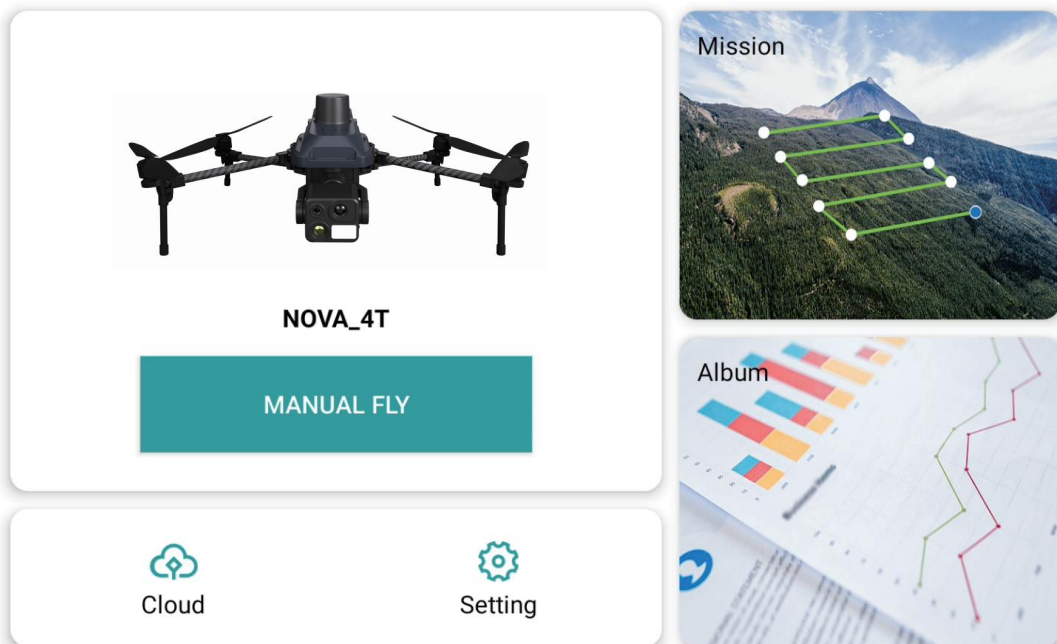
- | | | |
|------------------------|------------------------------------|--------------------|
| 16.Split-Screen Button | 18. Temperature Measurement Button | 20. Settings Panel |
| 17. Tracking Button | 19. Thermal imaging palette switch | |

Pre-Flight Check

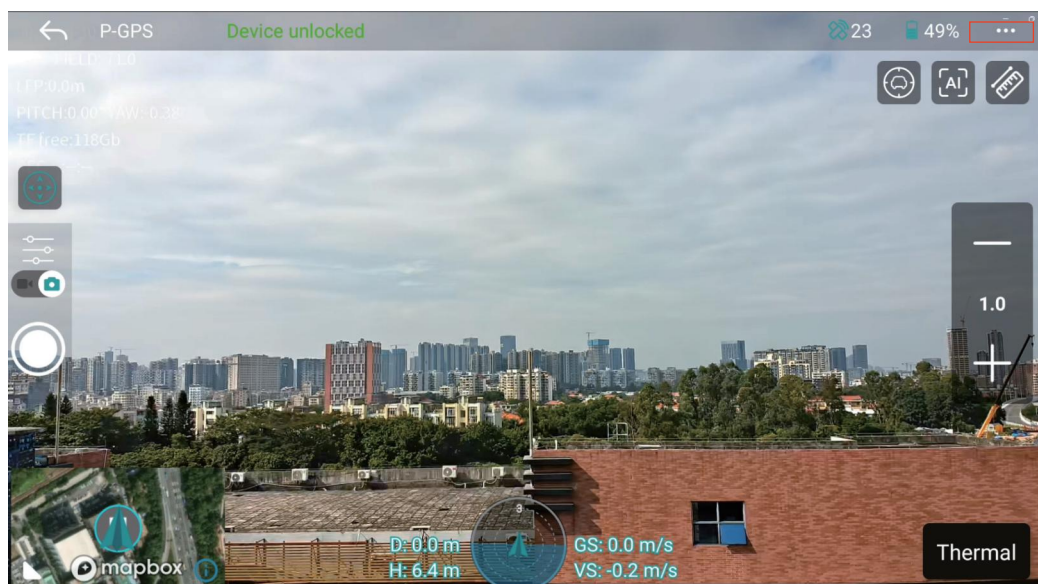
1. Ensure that the remote controller and the drone's intelligent battery have sufficient power.
2. Verify that the front and rear arms are fully extended.
3. Confirm that the propellers are fully deployed.
4. After powering on, listen for the motor self-test sound.
5. After powering on, check that the gimbal camera is functioning normally. (Gimbal camera startup time: 90 seconds.)
6. Verify that the GCS Fly app is operating normally.
7. Ensure the drone's intelligent battery is securely installed with the latches fully engaged.

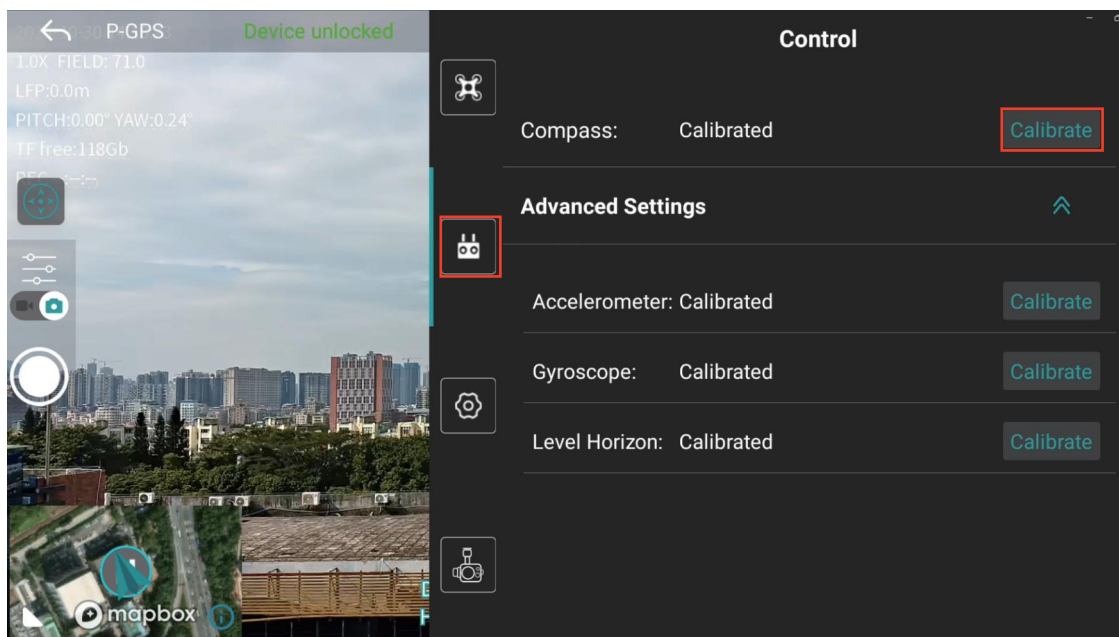
Calibrate the Magnetic Compass

Launch the GCS Fly app and navigate to the Manual Flight interface.



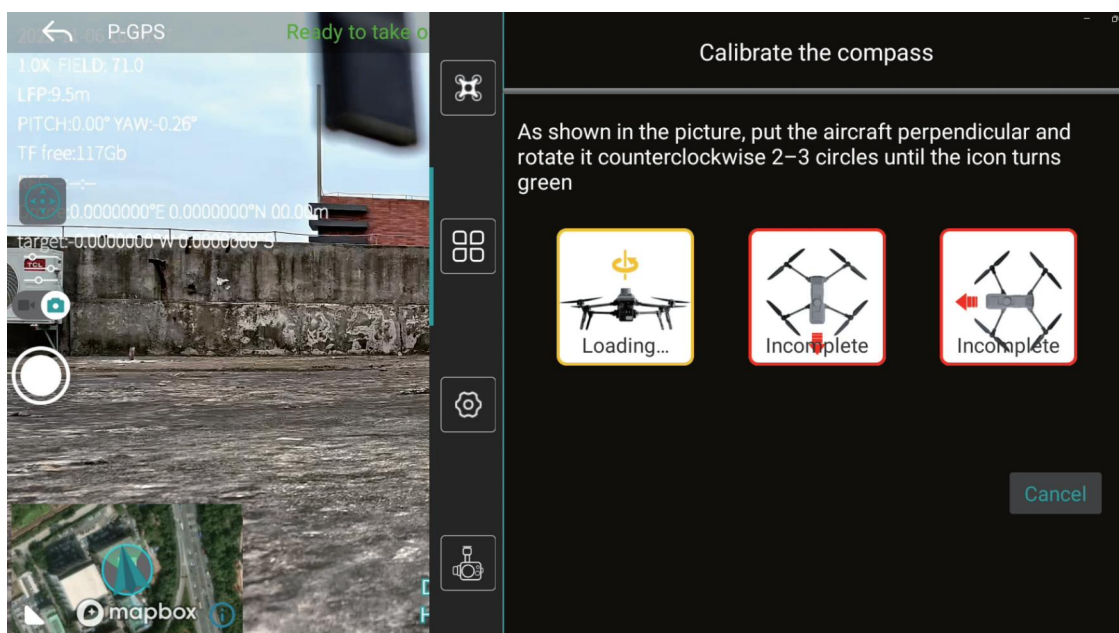
To calibrate the magnetic compass, tap the settings icon in the upper-right corner, select the remote control icon, enter the magnetic compass calibration interface, and tap Calibrate.



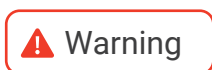


Warning

Before calibrating the drone, ensure that the environment is free from strong magnetic interference.



1. Position the aircraft horizontally on the ground in an open area until the calibration icon on the remote controller turns yellow.
2. Rotate the aircraft horizontally counterclockwise two to three full turns until the icon turns green, which indicates successful calibration.
3. Repeat these steps to position the drone according to the angles in the diagram before proceeding to the next calibration step.



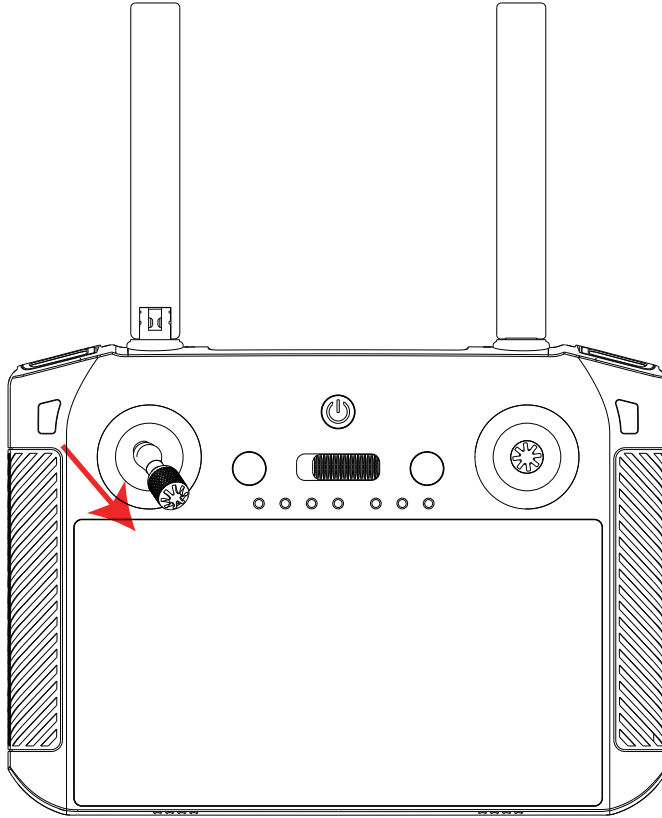
Warning

1. The first calibration maneuver is level flight.
2. The second is a nose-down vertical descent.
3. The third is a nose-forward level flight with the left wing vertical.

Manual Motor Start/Stop

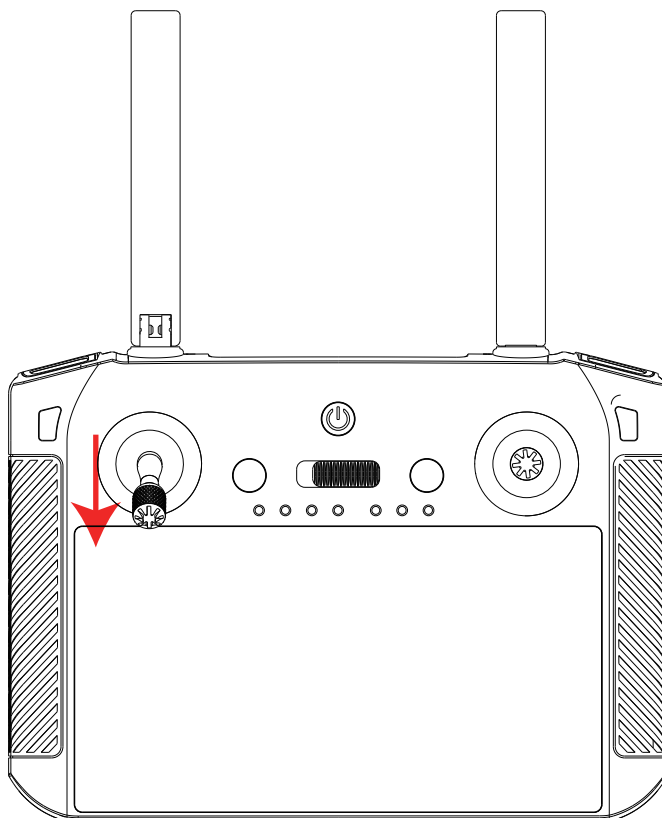
Starting the Motor

Pull the lever to start the motor. Release the lever immediately once the motor begins rotating.



Stop the Motor

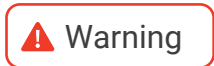
After the motor starts rotating, pull the lever. Release the lever after the motor has stopped for two seconds.



Basic Flight

Flight Steps

1. Place the drone on a flat, open surface with the user facing the tail.
2. Turn on the remote controller and the drone.
3. Launch the GCS Fly app to access the flight interface.
4. Wait for the camera to initialize. Once the app shows no error alerts, activate the motors.
5. Slowly push the throttle stick upward to enable a smooth takeoff.
6. Lower the throttle stick to descend the drone.
7. After descending smoothly, pull the throttle stick to its lowest position and hold it there for at least two seconds, until the motors stop.
8. After landing, power off the drone, followed by the remote controller.



1. Place the drone on a flat, stable surface before taking off.
2. Handheld takeoffs and landings are not supported.

Maintenance and Care

Drone Maintenance

1. Before each flight, carefully inspect the propellers for cracks, nicks, deformation, or wear. It is recommended that you inspect them again after each flight. Minor propeller damage may cause vibrations during flight and affect stability. If the damage is severe, replace the propellers immediately.
2. Before and after each flight, inspect the motor by listening for abnormal sounds and feeling for vibrations. Unusual friction or stuttering is typically caused by bearing wear or motor mount issues.
3. Before and after each flight, inspect the smart battery for swelling or leakage. If abnormalities are detected, cease use immediately and dispose of the battery safely according to regulations.
4. When storing the battery long-term, maintain a charge level between 50% and 60%. Store it in a cool, dry place between 15°C and 25°C, away from heat sources such as radiators or direct sunlight.
5. Clean the gimbal with cleaning agents and lens cloths designed for this purpose. Avoid corrosive chemicals.

Remote Control Maintenance

1. Avoid dropping the remote control or exposing it to moisture or high temperatures (e.g., leaving it inside a vehicle). Store it in a dry, well-ventilated place.
2. Calibrate the remote control every three months. Use the ground station software to observe channel response and ensure an accurate joystick travel range (e.g., prevent reverse movement in the throttle or steering channels).

Battery Safety Guidelines

To ensure the safe use of batteries and prevent accidents caused by improper operation, strictly adhere to the following precautions:

1. **Waterproofing and Moisture Protection:** Never expose batteries to liquids. Do not use them in the rain or in humid environments. If a battery comes into contact with liquid, immediately remove it and place it in a dry area away from flammable materials. Contact technical support.

2. **Do not use damaged batteries.**

Never use batteries that are bulging, leaking, or have damaged packaging.

3. **Temperature Environment**

Use batteries only within the specified temperature range. Excessive heat may cause a fire or an explosion, and extreme cold reduces battery performance.

4. **Do not disassemble or pierce.**

Never disassemble or puncture batteries with sharp objects.

5. **Leak Response**

Battery fluid is highly corrosive. If leakage occurs, move away immediately. If the fluid comes into contact with your skin or eyes, rinse the affected area thoroughly with water and seek medical attention.

6. **Storage and Impact Protection**

Store batteries out of the reach of children. Do not reuse batteries if they have been dropped or subjected to impact.

7. **Charge after cooling**

After flight, the battery remains hot. Allow it to cool to room temperature before charging; otherwise, it may fail to charge.

8. **Charging Temperature Range**

Charge the battery between 5°C and 45°C. The ideal charging temperature range is 22°C to 28°C, which extends the battery's life.

9. **Keep away from heat sources.** Do not place the battery near sources of heat, such as direct sunlight, hot car interiors, open flames, or heating furnaces.

10. **Avoid deep discharge storage.** Do not store the battery for extended periods after deep discharge, as this may cause irreversible cell damage.

11. **Protect from Static Electricity and Magnetic Fields**

Do not operate the aircraft in environments with strong static electricity (e.g., thunderstorms) or intense magnetic fields.

12. **Prevent short circuits.** Do not place batteries on conductive surfaces (e.g., metal shelves) or short-circuit terminals with metal objects.

13. **Use stable power sources for charging.**

Always charge using a voltage-stabilized power source. Unstable power supplies (e.g., generators or outdoor power stations) may damage batteries.

14. **Maintain a safe charging environment.** During charging, place the batteries and charging equipment on a flat, nonflammable surface (e.g., a concrete floor).

15. **Use official equipment.**

The use of non-official batteries and charging devices is strictly prohibited. If replacement is necessary, please purchase through official channels. Users are fully responsible for any malfunctions or accidents resulting from the use of non-official equipment.